DHARMSINH DESAI UNIVERSITY

Faculty of Management and Information Science

Project Title : PG and Hostel Management System

System Name : EasyPG Manager

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for *EasyPG Manager*, which is a web-based system designed to manage Paying Guest (PG) accommodations and hostels. This document provides a detailed description of the system's functionalities, workflows, user roles, and security requirements. It serves as a guide for developers who will build the system, designers who will create the user interface, and evaluators who will test the system against the requirements.

The system aims to solve the common challenges faced in PG management, such as unverified property listings, lack of transparency in payments, difficulty in tracking tenant records, and inefficient communication between tenants, owners, and administrators. By integrating OTP verification, CAPTCHA protection, secure payment gateways, and document validation, EasyPG Manager ensures that the platform is reliable, transparent, and secure for all stakeholders.

1.2 Scope

The scope of EasyPG Manager is to provide a centralized platform where students and working professionals can search for PG accommodations, book rooms, make payments, and manage their stay. PG owners can list their properties, manage tenants,

and track payments, while administrators oversee the system to ensure compliance and security.

The system covers the following areas:

- **Tenant Services**: Searching PGs, booking rooms, uploading ID proofs, making payments, and managing stay records.
- **Owner Services**: Registering properties, uploading verification documents, managing rooms and tenants, and monitoring payments.
- **Admin Services**: Verifying property documents, managing users, monitoring bookings and payments, handling complaints, and generating reports.

The system ensures transparency by requiring property verification before listings are made public. It ensures security by implementing OTP and CAPTCHA during registration and login. It ensures efficiency by automating notifications for payments, bookings, and due dates.

2. Number of Roles

The system defines three distinct roles, each with specific responsibilities and access rights:

1. **Student / Tenant** – End users who book and stay in PG accommodations.
2. **PG Owner** – Property owners who list and manage PGs.
3. **Admin** – System administrators who oversee operations, verify documents, and ensure compliance.

3. Detailed Role-Based Requirements

3.1 Student / Tenant Role

A. Registration and Login

Students or tenants begin by visiting the EasyPG Manager website. They select the option to register as a tenant and provide personal information such as their full name, email address, mobile number, residential address, city, pincode, and password. Once the details are submitted, the system sends an OTP to the registered mobile number or email address. The tenant must enter this OTP correctly to verify their identity. Additionally, CAPTCHA verification ensures that the registration is performed by a human and not an automated bot.

Only after successful OTP and CAPTCHA verification is the tenant's account activated. The tenant can then log in using their credentials and is redirected to the Student Dashboard. This process ensures that only genuine users gain access to the system, thereby preventing unauthorized access and fake registrations.

B. Browsing and Booking PGs

Once logged in, tenants can browse PG listings through the Student Dashboard. The system provides filters such as room type (single or double), availability, rent range, and amenities (Wi-Fi, meals, laundry, etc.). These filters allow tenants to quickly find PGs that match their preferences and budget.

When a tenant selects a PG and clicks *Book Now*, they are prompted to choose the room type, move-in date, and roommate preferences. The tenant must also upload a valid ID proof to complete the booking request. Depending on the owner's settings, the booking may be auto-approved or may require manual approval by the owner. If auto-approval is enabled, the booking is confirmed instantly. If owner approval is required, the booking status is marked as *Pending* until the owner reviews and accepts it.

This workflow ensures flexibility for tenants while giving owners control over who stays in their property.

C. Payment Handling

Tenants can pay rent through the integrated payment gateway, which supports UPI, net banking, and debit/credit cards. Each payment is recorded by the system with details such as the payment date and time, transaction ID, amount paid, and payment status (Paid, Due, or Overdue).

The system automatically sends notifications to both tenants and owners. Tenants receive payment confirmations and reminders for upcoming or overdue rent. Owners receive alerts when rent has been paid. This automation ensures that both parties are informed and reduces the chances of disputes.

D. Dashboard Functionalities

The Student Dashboard provides tenants with several modules:

- **My Bookings**: Displays active, pending, and past bookings.
- **Roommates**: Shows roommate names, contact details, and preferences.
- **Leave Requests**: Allows tenants to apply for temporary leave.
- **Move-Out Requests**: Enables tenants to submit planned move-out dates.
- **Past Records**: Provides access to previous PG stays and payment history.
- **Notifications**: Displays alerts for rent due, overdue payments, and booking updates.
- **Profile Management**: Allows tenants to update personal and address details.

This dashboard ensures that tenants can manage all aspects of their stay in one place, making the system user-friendly and efficient.

3.2 PG Owner Role

A. Registration and Login

PG owners register by selecting the *PG Owner* option during registration. They provide details such as name, email, mobile number, address, city, pincode, and password. Similar to tenants, OTP and CAPTCHA verification are required to activate the account. Once verified, owners are redirected to the Owner Dashboard.

B. Property Verification

Owners must upload property-related documents to verify their listings. These include identity proof (Aadhaar or PAN), property ownership proof (sale deed or property tax receipt), rental or lease agreement (if applicable), no objection certificate (if required), address proof (electricity or water bill), and bank or UPI details for rent transactions.

The admin reviews these documents and either approves or rejects them. Only verified properties are displayed to tenants. This process ensures that tenants only see legitimate PGs, thereby preventing fraud.

C. PG and Room Management

Owners can add and update PG details such as room types, rent, and amenities. They can track room availability and tenant allocation, approve or reject tenant bookings and room change requests, and monitor move-in and move-out dates. Owners can also manage offers, discounts, and promotions to attract tenants.

This functionality gives owners complete control over their properties and tenants, ensuring smooth operations.

D. Payment and Notifications

Owners can view tenant payments, due dates, and overdue alerts. They receive notifications for new bookings, payments, and cancellations. This ensures that owners are always informed about the financial and operational status of their PGs.

3.3 Admin Role

A. Login

Admins log in through a secure Admin Panel using a username and password. Once authenticated, they are redirected to the Admin Dashboard.

B. Responsibilities

Admins have several responsibilities:

- **Property Verification**: Approving or rejecting PG owner documents.
- **User Management**: Adding, blocking, deleting, or updating tenant and owner accounts.
- **PG and Booking Monitoring**: Viewing and tracking all listings, bookings, payments, and due dates.
- **Complaint Handling**: Resolving issues raised by tenants or owners.
- **Broadcast Notifications**: Sending system-wide alerts and announcements.
- **Report Generation**: Creating reports on bookings, payments, occupancy rates, and tenant/owner activity.

This role ensures that the system remains secure, transparent, and well-managed.

4. **Property Verification Documents (PG Owner)**

For a PG owner to list their property on *EasyPG Manager*, the system requires only two essential documents. The first document is an **identity proof**, which can be the owner's Aadhaar card. This ensures that the person registering is genuine and can be uniquely identified. Aadhaar is chosen because it is widely accepted across India and provides a reliable way to confirm the owner's identity.

The second document is a **property ownership proof**, which can be the property deed (sale deed) or, in cases where the deed is not readily available, a property tax receipt. This document establishes that the PG owner has legal rights over the property they are listing. By requiring this proof, the system ensures that only legitimate properties are displayed to tenants, preventing fraudulent or unauthorized listings.

The verification process is straightforward: the owner uploads these two documents through the Owner Dashboard. The administrator then reviews the submissions. If the documents are valid, the property is approved and becomes visible to tenants. If the documents are invalid or incomplete, the administrator rejects the submission, and the owner is notified to re-upload correct documents.

This simple two-step verification—identity proof plus property proof—keeps the process easy for owners while maintaining trust and transparency for tenants.
Workflow: Owner uploads documents → Admin reviews → Approve/Reject → PG listing becomes active if approved.

5. **Conclusion**

EasyPG Manager is a comprehensive, secure, and user-friendly platform for PG accommodation management. It provides tenants with seamless booking, payment, and stay management. It allows owners to manage verified property listings, tenants, and payments. It enables administrators to oversee the system, verify documents,

handle complaints, and generate reports. By combining role-based workflows, secure authentication, automated notifications, and property verification, EasyPG Manager ensures transparency, efficiency, and trustworthiness in PG and hostel management.